04 Network and QoS

Pro tips
? Mark audio EF / DSCP 46, video AF41 / 34, app sharing AF21 / 18, and signaling CS5 / 40. The mark has to survive the trust boundary.
? A short path and enough bandwidth beat a clever queue. A VPN is not a media design.
? G.711 at 20 ms is about 80 kbit/s at layer 3, and about 87.2 kbit/s with an 18-byte Ethernet header.

Interview traps
? Marking DSCP only on the PC and assuming the WAN still honors it.
? Treating media bypass as a bandwidth feature. Bypass changes where RTP goes, not the codec math.

Cheat sheet

Audio	DSCP 46	EF
Video	DSCP 34	AF41
Sharing	DSCP 18	AF21
Signaling	DSCP 40	CS5

Confirm the current port ranges on the network article you operate.
VPN pools are a bad place for real-time media.

Facts in the long lessons were checked 14 October 2026. This sheet is a memory aid.

CypherMonkey ? https://cyphermonkeylabs.com ? https://github.com/cipher0x9 ? https://github.com/CypherMonkeyLabs ? https://github.com/cipher0x9/microsoft-voice-cc-lab-free-university-mesmerizing ? https://github.com/CypherMonkeyLabs/microsoft-voice-cc-lab-free-university-mesmerizing ? https://x.com/Cypher0x9 ? https://huggingface.co/Cypher0x9 ?
(c) 2026 CypherMonkeyLabs. Independent educational project, not affiliated with or endorsed by any vendor. Trademarks belong to their owners. MIT code, CC BY-NC-SA content. No email.